



# ENGIN 604 INTRODUCTION TO PYTHON FOR FINANCE

## VARIABLES, CONVENTIONS & DATA TYPES

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## 1 Variables

A variable in Python is a space in your computer's RAM where you can store results and objects. To create a variable we use an "assignment", that is, an expression with an equality. What is to the left of the equals sign will be the name of the variable where the value is stored, and what is to the right of the equals sign is the value to be stored (*Aplicaciones de Manejo de Datos, 2020*).

For example, let's create the variable `phone_number` and store the value 12345678 in it:

```
phone_number = 12345678 # the symbol = is the assignment operator
phone_number
```

```
## 12345678
```

In python it is not necessary to end a line of code with ; (semicolons), but they do serve a purpose:

```
a = 1; b = 2; c = 3
```

You can also assign values to several variables in a single line of code:

```
a, b, c = 1, 2, 3
```

## 2 Conventions

1. There are reserved variable names that must not be used because they are reserved by Python:

```
# library
import keyword

# reserved keywords
keyword.kwlist
```

2. Single-line comments are written with #.

```
# this is a single-line comment
```

3. Multi-line comments start with ''' and end with '''.

```
"""
this is
a
multi-line
```

*comment*  
"""

### 3 Data Types (scalars)

1. What are the data types in Python? How can I identify them and convert them into other data types?

#### 3.1 Numeric Expressions

1. What numeric operations can I apply?
2. What is the result of:

$$1 + 2^3/4 \times 5$$

3. Compute the present value of a cash flow of \$1000 over a total of 5 periods. Assume a discount rate of 10%. The present value (PV) is equal to:

$$PV = \sum_{t=0}^T \frac{F_t}{(1+r)^t}$$

#### 3.2 Strings

1. Given the following text: “*There are some things money can’t buy, for everything else there’s mastercard*”. Answer the following:
  - a. How many characters does the text contain?
  - b. How many times does the letter r appear? How many times does the letter r appear after the “,”?
  - c. What position is “mastercard” at?
  - d. Replace “mastercard” with “Mastercard”.
2. Write your full name in lowercase, preserving the spaces.
  - a. Print your full name to the screen in uppercase. (e.g. GUIDO VON ROSSUM)
  - b. Print your full name to the screen with only the first letters in uppercase. (e.g. Guido Von Rossum)
3. Apply the following format<sup>1</sup> to the text: ‘2.45535 dollars is equivalent to \$CLP1808.12’
  - a. From ‘2.45535’ to 2.45 (two decimal places)
  - b. ‘dollars’ as text
  - c. From ‘1808.12’ to 1808 (integer)
4. What is ‘2’ + ‘25’?

#### 3.3 Boolean Logic

1. What logical operations can I apply?
2. Given the following text: “*There are some things money can’t buy, for everything else there’s mastercard*”. Answer the following:
  - a. Is the letter ‘z’ present?
  - b. Is the letter ‘m’ present?
  - c. Is the word ‘visa’ present?
3. Show that De Morgan’s law holds.

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<sup>1</sup><https://docs.python.org/3/library/stdtypes.html#printf-style-string-formatting>

## 4 Appendix

Table 1: De Morgan's Law

| A     | B     | A and B | not(A and B)      | not A or not B                  |
|-------|-------|---------|-------------------|---------------------------------|
| True  | False | False   | not(False) = True | not(True) or not(False) = True  |
| False | True  | False   | not(False) = True | not(False) or not(True) = True  |
| True  | True  | True    | not(True) = False | not(True) or not(True) = False  |
| False | False | False   | not(False) = True | not(False) or not(False) = True |